
ASSET PRICING AND RISK

Cheat Sheet

Authors

Srikar
(in Association with Alex)

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WEEK 1

1.1 PRINCIPAL REDUCTION

Used when a borrower pays more than the interest due in order to reduce the loan balance.

$$\text{Remaining Principal} = \text{Initial Principal} \times (1 - \text{reduction rate})^T$$

1.2 NET PRESENT VALUE

Net Present Value (NPV) is used to value projects by discounting future cash flows to time 0.

$$V_0 = CF_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \dots$$

1.3 FUTURE VALUE

FUTURE VALUE FORMULA

Future Value (FV) calculates the value of an investment at a future date given a specific interest rate.

$$FV = PV \cdot (1+r)^T$$

FUTURE VALUE OF AN ANNUITY

Future Value of an Annuity (FVA) is used to find the accumulated value of a series of deposits at the end of a term.

$$FVA(r, n) = \frac{1}{r}((1+r)^n - 1)$$

1.4 EFFECTIVE ANNUAL RATE

EFFECTIVE ANNUAL RATE (EAR)

Used to compare interest rates with different compounding frequencies (n periods per year).

$$EAR = \left(1 + \frac{r}{n}\right)^n - 1$$

IMPLIED EAR FOR SHORT-TERM LOANS

A specific adaptation of the EAR formula for loans lasting D days (e.g. payday loans).

$$EAR = \left(\frac{FV}{PV}\right)^{365/D} - 1$$

1.5 PRESENT VALUE AND ANNUITIES

PRESENT VALUE OF AN ANNUITY

Present Value of an Annuity (PVA) is used for fixed payments over a set duration, such as mortgages or retirement withdrawals.

$$PV = C \times ADF(r, T), \quad ADF(r, T) = \frac{1}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

INTERNATIONAL NET PRESENT VALUE

International NPV (Discount-then-Convert) values foreign projects by discounting in the local currency and then converting to the domestic currency at the spot exchange rate.

$$NPV_{\text{domestic}} = NPV_{\text{foreign}} \times \text{Spot Rate}$$

1.6 GROWTH AND DISCOUNTING

PRESENT VALUE OF A GROWING PERPETUITY

Used for cash flows that continue indefinitely at a constant growth rate g .

$$V_0 = \frac{C}{r - g}$$

2.1 CREDIT RISK AND PORTFOLIO IMMUNIZATION

PROBABILITY OF REPAYMENT AND DEFAULT

Probability of Repayment (p_R) is computed when the recovery rate R , the promised yield r_p , and the risk-free rate r_f are known:

$$p_R = \frac{\frac{1 + r_f}{1 + r_p} - R}{1 - R}.$$

The corresponding **Probability of Default** (p_D) is:

$$p_D = 1 - p_R.$$

DURATION MATCHING FOR A PORTFOLIO

To immunize a portfolio against interest rate changes, the value-weighted durations must match the target duration:

$$V_1 D_1 + V_2 D_2 = D_{\text{target}} (V_1 + V_2).$$

CREDIT SPREAD

The **Credit Spread** is defined as the yield difference between a corporate bond and a government bond of the same maturity:

$$\text{Credit Spread} = \text{Corporate Bond Yield} - \text{Government Bond Yield}.$$

2.2 BOND RATES OF RETURN

GENERAL RATE OF RETURN

The rate of return measures the actual gain or loss over a specific holding period:

$$R = \frac{\text{Coupon Income} + \text{Price Change}}{\text{Initial Investment}} = \frac{P_1 + C - P_0}{P_0}.$$

2.3 BOND PRICING AND YIELD TO MATURITY

STANDARD COUPON BOND PRICING (ANNUAL)

For an annual coupon bond, the price is:

$$P_0 = \sum_{t=1}^T \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T}.$$

ANNUITY REPRESENTATION OF COUPON BONDS

Using the present value of an annuity for coupons:

$$P_0 = \frac{C}{y} \left[1 - \frac{1}{(1+y)^T} \right] + \frac{F}{(1+y)^T}.$$

SEMI-ANNUAL COUPON PRICING

For semi-annual coupon bonds, the coupon is $C/2$, the yield is $y/2$, and the total number of periods is $2T$.

PRICING VIA SPOT RATES (BOOTSTRAPPING)

Bond prices can also be computed using spot rates:

$$P = \frac{C_1}{1+r_{0,1}} + \frac{C_2}{(1+r_{0,2})^2} + \cdots + \frac{C_T + F}{(1+r_{0,T})^T}.$$

2.4 FORWARD RATES AND ARBITRAGE

FORWARD RATES

Forward rates represent current forecasts of future spot rates.

GENERAL FORWARD RATE FORMULA

The forward rate between t_1 and t_2 is:

$$f_{t_1, t_2} = \left[\frac{(1+r_{0, t_2})^{t_2}}{(1+r_{0, t_1})^{t_1}} \right]^{\frac{1}{t_2-t_1}} - 1.$$

ONE-YEAR FORWARD RATE (SIMPLIFIED)

For a one-year forward rate:

$$1 + f_t = \frac{(1+r_{0, t})^t}{(1+r_{0, t-1})^{t-1}}.$$

2.5 MEASURES OF INTEREST-RATE RISK

MACAULAY DURATION

Macauley Duration (D) is defined as:

$$D = \frac{\sum_{t=1}^T t \cdot PV(CF_t)}{P}.$$

MODIFIED DURATION

Modified Duration (D_m) adjusts Macauley duration for yield:

$$D_m = \frac{D}{1+y}.$$

DURATION APPROXIMATION OF PRICE CHANGES

The first-order (duration) approximation of percentage price change is:

$$\frac{\Delta P}{P} \approx -D_m \Delta y.$$

CONVEXITY ADJUSTMENT

Including convexity yields the second-order approximation:

$$\frac{\Delta P}{P} \approx -D_m \Delta y + \frac{\text{Con} (\Delta y)^2}{2}.$$

3.1 BASIC STATISTICS AND WEALTH CALCULATIONS

EXPECTED PAY-OFF

The expected pay-off of a random variable X is:

$$E[X] = \sum_{i=1}^n p_i x_i.$$

VARIANCE

The variance of X is defined as:

$$\text{Var}(X) = \sum_{i=1}^n p_i (x_i - E[X])^2.$$

In the lecture notes, variance is denoted by ω^2 and, for fair lotteries, is associated with $E[\tilde{\epsilon}^2]$.

EXPECTED FINAL WEALTH

Expected terminal wealth is given by:

$$E[W_T] = W_0 + E[X].$$

TIME-SCALING OF VOLATILITY

Expected return scales linearly with time, while volatility scales with the square root of time:

$$\sigma_t = \sigma\sqrt{t}.$$

VARIANCE OF A BERNOULLI-STYLE LOTTERY

For a lottery Z with outcomes H and $-L$,

$$\text{Var}(Z) = p(1-p)(H+L)^2.$$

3.2 EXPECTED UTILITY THEORY

EXPECTED UTILITY (DISCRETE)

Expected utility over discrete outcomes is:

$$V = E[U(W)] = \sum_{i=1}^n p_i U(W_i).$$

CRRA UTILITY FUNCTIONS

The lecture notes define power utility as:

$$u(W) = \frac{1}{\sigma} W^\sigma.$$

The exercise solutions use the standard CRRA form:

$$u(W) = \begin{cases} \frac{W^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \ln(W), & \gamma = 1. \end{cases}$$

LINEAR TRANSFORMATION OF UTILITY

Expected utility is unique up to a linear monotonic transformation:

$$W(P) = a + bV(P).$$

3.3 RISK MEASURES: CERTAINTY EQUIVALENT AND RISK PREMIUM

CERTAINTY EQUIVALENT

The certainty equivalent CE is defined implicitly by:

$$u(CE) = E[u(W)].$$

CERTAINTY EQUIVALENT FOR CRRA UTILITY

Explicit expressions used in Question 2 are:

$$CE = \begin{cases} [(1-\gamma)V]^{\frac{1}{1-\gamma}}, & \gamma \neq 1, \\ e^V, & \gamma = 1. \end{cases}$$

These explicit inverse functions are not provided in the lecture notes, which define the certainty equivalent only implicitly.

RISK PREMIUM

The Markowitz risk premium π is defined implicitly by:

$$u(E[X] - \pi) = E[u(X)].$$

EXPLICIT RISK PREMIUM FOR CRRA

Derived in Question 6, the explicit expression is:

$$\pi = E(X) - (E[X^{1-\gamma}])^{\frac{1}{1-\gamma}}.$$

MEASURES OF RISK AVERSION

The Pratt–Arrow measure of absolute risk aversion is:

$$R(W) = -\frac{u''(W)}{u'(W)}.$$

Relative risk aversion is defined as:

$$R_r(W) = WR(W).$$

3.4 PROSPECT THEORY

INDEPENDENCE AXIOM

Preferences satisfy the independence axiom if:

$$P \succ Q \iff \alpha P + (1 - \alpha)R \succ \alpha Q + (1 - \alpha)R.$$

OVERALL VALUE IN PROSPECT THEORY

The overall value of a prospect is:

$$V(P) = \sum_i \pi_i v(x_i),$$

where π_i are decision weights and $v(\cdot)$ is the value function.

PROSPECT THEORY VALUE FUNCTION

For gains, the value function is:

$$v(x) = x^\alpha,$$

with the solutions using $\alpha = 0.88$.

PROBABILITY WEIGHTING FUNCTION

The probability weighting function for gains is:

$$w^+(p) = \frac{p^\delta}{(p^\delta + (1 - p)^\delta)^{1/\delta}},$$

4.1 PORTFOLIO OPTIMIZATION

LAGRANGIAN FORMULATION

The Lagrangian used to minimize portfolio variance subject to a return constraint is

$$L(w, \lambda) = \frac{1}{2} w' V w - \lambda (w' (\bar{R} - R_f e) - (\bar{R}_p - R_f)).$$

RELATIONSHIP BETWEEN λ AND γ

Without a risk-free asset, the portfolio weights are solved using

$$\lambda = \frac{\delta \bar{R}_p - \alpha}{\Delta}$$

and

$$\gamma = \frac{\zeta - \alpha \bar{R}_p}{\Delta},$$

which follow from the first-order conditions of the Markowitz optimization problem.

4.2 BASIC PORTFOLIO STATISTICS AND THE SHARPE RATIO

EXPECTED RETURN

For a two-asset portfolio consisting of a risky asset and a risk-free asset,

$$E[R_p] = w_{\text{risky}} E[R_{\text{risky}}] + w_{\text{rf}} R_f.$$

STANDARD DEVIATION

The standard deviation of a two-asset portfolio with one risky asset is

$$\sigma_p = w_{\text{risky}} \sigma_{\text{risky}}.$$

OVERALL PORTFOLIO WEIGHTS

The weight of a specific stock in an overall portfolio that includes T-bills is

$$\text{Weight} = (\text{weight of risky fund in overall portfolio}) \times (\text{weight of stock in risky fund}).$$

SHARPE RATIO

The Sharpe ratio of asset i is defined as

$$SR = \frac{\bar{R}_i - R_f}{\sigma_i}.$$

4.3 TWO-ASSET PORTFOLIOS AND EQUILIBRIUM

PERFECT NEGATIVE CORRELATION

With perfect negative correlation $\rho = -1$, the portfolio standard deviation is

$$\sigma_P = |w_A \sigma_A - w_B \sigma_B|.$$

VARIANCE AND COVARIANCE

Variance is given by

$$\sigma^2 = (\sigma)^2,$$

while covariance between assets X and Y is

$$\sigma_{X,Y} = \rho \sigma_X \sigma_Y.$$

4.4 EFFICIENT FRONTIER AND MATRIX ALGEBRA

In the n-asset case, V denotes the covariance matrix and $\bar{R}$ the vector of expected returns.

EFFICIENT SET CONSTANTS

The constants characterizing the mean–variance frontier are

$$\alpha = e'V^{-1}\bar{R},$$

$$\delta = e'V^{-1}e,$$

$$\zeta = \bar{R}'V^{-1}\bar{R},$$

and

$$\Delta = \zeta\delta - \alpha^2.$$

INVERSE OF A 2×2 COVARIANCE MATRIX

For two assets, the inverse covariance matrix is

$$V^{-1} = \frac{1}{\det(V)} \begin{pmatrix} \sigma_Y^2 & -\sigma_{X,Y} \\ -\sigma_{X,Y} & \sigma_X^2 \end{pmatrix},$$

where

$$\det(V) = \sigma_X^2 \sigma_Y^2 - (\sigma_{X,Y})^2.$$

WEIGHTS OF EFFICIENT PORTFOLIOS

Efficient portfolio weights satisfy

$$w^* = \lambda V^{-1}\bar{R} + \gamma V^{-1}e.$$

PORTFOLIO VARIANCE IDENTITY

For efficient portfolios, portfolio variance can be written as

$$\sigma_P^2 = \gamma + \lambda \bar{R}_P.$$

4.5 SPECIFIC PORTFOLIO TYPES

GLOBAL MINIMUM VARIANCE PORTFOLIO

The GMV portfolio has weights

$$w_{\text{GMV}} = \frac{1}{\delta} V^{-1} e,$$

expected return

$$\bar{R}_{\text{GMV}} = \frac{\alpha}{\delta},$$

and volatility

$$\sigma_{\text{GMV}} = \frac{1}{\sqrt{\delta}}.$$

TANGENCY PORTFOLIO

The tangency (maximum Sharpe ratio) portfolio has expected return

$$\bar{R}_{\text{TAN}} = \frac{\alpha R_f - \zeta}{\delta R_f - \alpha}.$$

With a risk-free asset, the weights are

$$w^* = \lambda V^{-1} (\bar{R} - R_f e),$$

where

$$\lambda = \frac{\bar{R}_p - R_f}{\zeta - 2\alpha R_f + \delta R_f^2}.$$

4.6 UTILITY AND EFFICIENCY PROPERTIES

MARKOWITZ UTILITY

The Markowitz utility function is

$$U(R_p) = E[R_p] - \frac{\gamma}{2} \text{Var}(R_p).$$

OPTIMAL PORTFOLIO UNDER RISK AVERSION

For risk aversion parameter ρ , the optimal coefficients are

$$\lambda = \frac{1}{\rho}$$

and

$$\gamma = \frac{1}{\delta} \left(1 - \frac{\alpha}{\rho} \right).$$

COVARIANCE WITH THE GMV PORTFOLIO

The covariance between any efficient portfolio and the GMV portfolio satisfies

$$\text{Cov}(R_{\text{mv}}, R_p) = \text{Var}(R_{\text{mv}}).$$

CORRELATION AND EFFICIENCY

For an efficient portfolio a and any portfolio b , the correlation is

$$\text{Corr}(R_a, R_b) = \frac{SR_b}{SR_a}.$$

5.1 CAPITAL ASSET PRICING MODEL

CAPM EXPECTED RETURN

The CAPM (Security Market Line) gives the fair or expected return of asset k as

$$E(r_k) = r_f + \beta_k(E[r_m] - r_f).$$

ASSET BETA

Asset beta measures sensitivity to market returns and is defined as

$$\beta_k = \frac{\text{Cov}(r_k, r_m)}{\text{Var}(r_m)} = \frac{\sigma_{km}}{\sigma_m^2}.$$

BETA AND CORRELATION

Beta can also be expressed using correlation and standard deviations as

$$\beta_k = \rho_{km} \frac{\sigma_k}{\sigma_m}.$$

TOTAL RISK DECOMPOSITION

Total variance is the sum of systematic and idiosyncratic components:

$$\sigma_k^2 = \beta_k^2 \sigma_m^2 + \sigma_{\epsilon k}^2.$$

5.2 EFFICIENT PORTFOLIOS AND THE CAPITAL MARKET LINE

CAPITAL MARKET LINE

The Capital Market Line relates expected return to total risk for efficient portfolios:

$$R_p = R_f + \frac{R_m - R_f}{\sigma_m} \sigma_p.$$

PORTFOLIO VARIANCE FOR LARGE PORTFOLIOS

For large portfolios, variance is approximated by

$$\sigma_p^2 = \frac{1}{n} \times \text{Average Variance} + \frac{n-1}{n} \times \text{Average Covariance}.$$

PORTFOLIO BETA

The beta of a portfolio is the weighted average of individual asset betas:

$$\beta_p = \sum_i w_i \beta_i.$$

5.3 PERFORMANCE MEASURES

ALPHA

Alpha measures abnormal performance relative to CAPM:

$$\alpha_k = \text{Actual Return} - [r_f + \beta_k(R_m - r_f)].$$

TREYNOR RATIO

The Treynor ratio evaluates excess return per unit of systematic risk:

$$\text{Treynor Ratio} = \frac{\mu_A - r_f}{\beta_A}.$$

5.4 PROJECT VALUATION

NET PRESENT VALUE

Net present value is used to evaluate project viability:

$$\text{NPV} = \sum_{t=1}^n \frac{CF_t}{(1+r)^t} - \text{Initial Investment}.$$

ANNUITY FACTOR

The present value of an annuity (PVA) is used as a numerical shortcut for constant cash flows in NPV calculations.

5.5 BASIC STATISTICAL CALCULATIONS

SCENARIO-BASED EXPECTED RETURN

When outcomes are given by discrete probability states,

$$\mu = \sum_i P_i R_i.$$

SCENARIO-BASED VARIANCE

Variance under discrete states is

$$\sigma^2 = \sum_i P_i (R_i - \mu)^2.$$

SCENARIO-BASED COVARIANCE

Covariance between asset X and the market M is

$$\text{Cov}(X, M) = \sum_i P_i (R_{X,i} - \mu_X)(R_{M,i} - \mu_M).$$

CORRELATION COEFFICIENT

The correlation between assets A and B is

$$\rho_{AB} = \frac{\text{Cov}(r_A, r_B)}{\sigma_A \sigma_B}.$$

5.6 UTILITY MAXIMIZATION

LAGRANGIAN FOR PORTFOLIO OPTIMIZATION

The Lagrangian used to derive first-order conditions for investor utility maximization is

$$L = \sum_i w_i \mu_i - \frac{1}{2\gamma_k} \sum_i \sum_j w_i w_j \sigma_{ij} + \lambda \left(1 - \sum_i w_i \right).$$

6.1 ARBITRAGE PRICING THEORY

APT EXPECTED RETURN

The general multi-factor APT expected return formula is

$$E[R_i] = \lambda_0 + \sum_{z=1}^k b_{iz}\lambda_z.$$

Here,

$$\lambda_0 = R_f,$$

b_{iz} = sensitivity (beta) of asset i to factor z,

and

λ_z = risk premium of factor z.

TWO-FACTOR APT FORMULA

In the two-factor case, expected portfolio return is

$$E[R_p] = R_f + b_{p1}\lambda_1 + b_{p2}\lambda_2.$$

6.2 ARBITRAGE PORTFOLIO CONSTRUCTION

ZERO-NET-INVESTMENT CONDITION

An arbitrage portfolio must satisfy

$$\sum_{i=1}^n w_i = 0.$$

ZERO-RISK (FACTOR NEUTRALITY) CONDITION

Factor neutrality requires

$$\sum_{i=1}^n w_i b_{zi} = 0 \quad \text{for all factors } z.$$

ARBITRAGE PROFIT CONDITION

The expected return on an arbitrage portfolio must be strictly positive:

$$E[R_p] = \sum_{i=1}^n w_i E[R_i] > 0.$$

ONE-FACTOR ARBITRAGE RATIO

In the one-factor case, mispricing can be identified by comparing

$$\frac{E[R_i] - R_f}{\beta_i}.$$

If this ratio differs across well-diversified portfolios, an arbitrage opportunity exists.

6.3 CAPM AND BETA DECOMPOSITION

CAPM RISK PREMIUM

Under the CAPM, the excess return of asset i is

$$E[R_i] - R_f = \beta_{Mi}(E[R_M] - R_f).$$

MARKET BETA VIA APT FACTORS

The market beta of a portfolio can be expressed using APT factor sensitivities as

$$\beta_{MA} = b_{1A}\beta_{MF1} + b_{2A}\beta_{MF2},$$

where

$$\beta_{MFz} = \text{market beta of factor } z.$$

6.4 FACTOR TRANSFORMATION AND ORTHOGONALIZATION

LINEAR FACTOR TRANSFORMATION

New factors can be defined as linear combinations of original factors:

$$g_1 = c_1 f_1 + c_2 f_2,$$

$$g_2 = c_3 f_1 + c_4 f_2.$$

TRANSFORMATION OF RISK PREMIA

Risk premia transform according to

$$\begin{bmatrix} \lambda_{g1} \\ \lambda_{g2} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} \begin{bmatrix} \lambda_{f1} \\ \lambda_{f2} \end{bmatrix}.$$

ORTHOGONALITY CONDITIONS

For orthogonal factors with unit variance, the coefficients must satisfy

$$c_1 c_3 + c_2 c_4 = 0,$$

$$c_1^2 + c_2^2 = 1,$$

and

$$c_3^2 + c_4^2 = 1.$$

6.5 ADDITIONAL NO-ARBITRAGE CONDITION

COVERED INTEREST PARITY

Covered interest parity implies the no-arbitrage condition

$$\frac{S_0}{F_0}(1 + r_0) = 1 + r_0^*,$$

where

S_0 = spot exchange rate,

F_0 = forward exchange rate,

r_0 = domestic interest rate,

and

r_0^* = foreign interest rate.

The CAPM can be interpreted as a single-factor model in which all systematic risk is captured by market movements, whereas the APT decomposes systematic risk into multiple underlying economic factors, allowing asset returns to be explained by several distinct sources of risk.

7.1 MARKET COMPLETENESS AND MATRIX ALGEBRA

To determine whether a market is *complete*, the following conditions are used.

DETERMINANT CONDITION (2×2 CASE)

For a market with assets A and B and states H and M, the payoff matrix X must be invertible. The determinant is

$$\det(X) = A_H B_M - A_M B_H \neq 0.$$

RANK CONDITION

A market is complete if and only if the row rank of the payoff matrix X equals the number of states k:

$$\text{rank}(X) = k.$$

REDUNDANCY

If the determinant is zero, assets are redundant. In this case, one asset can be written as a scalar multiple of another:

$$A = \delta B.$$

7.2 ARBITRAGE CONDITIONS

DEFINITION OF ARBITRAGE

An arbitrage opportunity exists if there is a portfolio θ such that

$$P'\theta \leq 0 \quad \text{and} \quad X\theta \geq 0,$$

with at least one inequality holding strictly.

POSITIVITY OF STATE PRICES

In a complete market, the market is arbitrage-free if and only if all Arrow–Debreu state prices are strictly positive:

$$q \gg 0.$$

ARBITRAGE-FREE PRICE RANGE

In the 2×2 case, assuming $\det(X) > 0$, the price of asset B must satisfy

$$\frac{B_M}{A_M} P_A > P_B > \frac{B_H}{A_H} P_A$$

to rule out arbitrage.

7.3 ARROW–DEBREU SECURITIES AND STATE PRICES

GENERAL STATE PRICE VECTOR

The vector of Arrow–Debreu prices is given by

$$q = (X')^{-1}P.$$

INVERSE MATRIX IN THE 2×2 CASE

For a 2×2 payoff matrix, the inverse is

$$X^{-1} = \frac{1}{\det X} \begin{pmatrix} B_M & -B_H \\ -A_M & A_H \end{pmatrix}.$$

INTERPRETATION OF AD PRICES

The Arrow–Debreu price of state j is the cost of a portfolio that pays

$$1 \text{ in state } j \text{ and } 0 \text{ in all other states.}$$

RISK-FREE ASSET PRICE

The price of a risk-free asset that pays 1 in every state is the sum of all state prices:

$$P_{\text{risk-free}} = \sum_i q_i = \frac{1}{1 + r_f}.$$

7.4 FIRM CLAIMS: DEBT, EQUITY, AND CDS

DEBT PAYOFF

The payoff of debt is

$$D = \min(A, F),$$

where A is the firm asset value and F is the face value of debt.

EQUITY PAYOFF

Equity payoff is

$$E = \max(A - F, 0),$$

which can also be written as

$$E = A - D.$$

CREDIT DEFAULT SWAP PAYOFF

The payoff of a CDS is

$$\text{Payoff}_{\text{CDS}} = \begin{cases} F - A_C, & \text{if } A_C < F, \\ 0, & \text{otherwise.} \end{cases}$$

FIRM VALUE DECOMPOSITION

The fair value of firm assets satisfies

$$p_A = p_D + p_E.$$

7.5 ADVANCED MATRIX SOLVING

SUB-MATRIX DETERMINANTS

For sparse 3×3 markets, define δ_k as the determinant of the 2×2 sub-matrix obtained by removing the last column and row k . For example,

$$\delta_2 = A_1 B_3 - A_3 B_1.$$

ARROW-DEBREU WEIGHTS IN THE SPARSE 3×3 CASE

The replication weights are given by

$$\theta_{3,C} = \frac{1}{C_3},$$
$$\theta_{1,C} = \frac{\delta_1}{\delta_3 C_3},$$

and

$$\theta_{2,C} = -\frac{\delta_2}{\delta_3 C_3}.$$

7.6 OPTIONS AND PUT-CALL PARITY

OPTION PRICING VIA STATE PRICES

The price of any derivative security is

$$\text{Price} = \sum_i (\text{Payoff}_i \times q_i).$$

PUT-CALL PARITY

Put-call parity implies

$$P + S = C + PV(K),$$

where P is the put price, C is the call price, S is the asset price, and $PV(K)$ is the present value of the strike price.

7.7 STOCHASTIC DISCOUNT FACTOR AND RISK-NEUTRAL PROBABILITIES

STOCHASTIC DISCOUNT FACTOR

The stochastic discount factor is

$$m_i = \frac{q_i}{\pi_i},$$

where π_i is the true probability of state i .

RISK-NEUTRAL PROBABILITIES

Risk-neutral probabilities are given by

$$\hat{\pi}_i = \frac{q_i}{\sum_j q_j}.$$

8.1 ASSET PRICE DYNAMICS AND STATISTICS

These formulas describe the behavior of the stock price S and its log returns Z in a binomial tree.

STOCK PRICE DYNAMICS

The stock price at step n after k up-moves is

$$S_{n,k} = U^k D^{n-k} S_0,$$

where

$$U = e^u, \quad D = e^d.$$

CONDITIONAL MOMENTS OF LOG RETURNS

The conditional expected log return over one step is

$$E_n[Z_{n+1}] = pu + (1-p)d.$$

The conditional variance is

$$\text{Var}(Z) = p(1-p)(u-d)^2.$$

Hence, the single-step volatility is

$$\text{StdDev}(Z) = (u-d)\sqrt{p(1-p)}.$$

Over n steps, volatility scales as

$$\text{Volatility}_n = \sqrt{n} \text{Volatility}_1.$$

UNCONDITIONAL DISTRIBUTION

The probability of reaching node (n, k) is

$$\Pr(S_n = S_{n,k}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

8.2 MODEL CALIBRATION

Assuming $p = \frac{1}{2}$ and a time step Δ , the parameters are chosen to match the mean and variance.

UP AND DOWN FACTORS

The up-move and down-move log factors are

$$u = \mu\Delta + \sigma\sqrt{\Delta}, \quad d = \mu\Delta - \sigma\sqrt{\Delta}.$$

RISK-FREE RETURN

The gross risk-free return over one period is

$$R = e^{r\Delta}.$$

8.3 RISK-NEUTRAL VALUATION AND REPLICATION

RISK-NEUTRAL PROBABILITY

The risk-neutral probability is

$$p^* = \frac{R - D}{U - D}.$$

BACKWARD INDUCTION

The value of a derivative at node (n, k) is given by

$$V_{n,k} = e^{-r\Delta} [p^* V_{n+1,k+1} + (1 - p^*) V_{n+1,k}].$$

REPLICATING PORTFOLIO

The number of shares of stock (delta) held is

$$\delta_{n,k} = \frac{V_{n+1,k+1} - V_{n+1,k}}{S_{n+1,k+1} - S_{n+1,k}}.$$

The number of bonds (alpha) is

$$\alpha_{n,k} = \frac{\delta_{n,k} S_{n+1,k+1} - V_{n+1,k+1}}{B_{n+1}}.$$

SELF-FINANCING CONDITION

A strategy is self-financing if

$$\delta_n S_{n+1} - \alpha_n B_{n+1} = \delta_{n+1} S_{n+1} - \alpha_{n+1} B_{n+1}.$$

8.4 OPTION-SPECIFIC FORMULAS

INTRINSIC VALUE

For a call option,

$$IV_{n,k} = S_{n,k} - X,$$

and for a put option,

$$IV_{n,k} = X - S_{n,k}.$$

EUROPEAN AND AMERICAN OPTIONS

For a European option,

$$V_{n,k} = CV_{n,k}.$$

For an American option,

$$V_{n,k} = \max\{IV_{n,k}, CV_{n,k}\}.$$

EUROPEAN OPTION VALUE (EXPECTATION METHOD)

The time-0 value of a European option is

$$V_0 = e^{-rT} \sum_{k=0}^n \binom{n}{k} (p^*)^k (1 - p^*)^{n-k} \cdot \text{Payoff}_{n,k}.$$

9.1 OPTION PAYOFFS AND PRICING

OPTION PAYOFFS AT MATURITY

For European options at expiration τ :

$$\begin{aligned}\text{European Call: } C_\tau &= \max(S_\tau - K, 0), \\ \text{European Put: } P_\tau &= \max(K - S_\tau, 0), \\ \text{Written (Short) Call: } & - \max(S_\tau - K, 0), \\ \text{Written (Short) Put: } & - \max(K - S_\tau, 0).\end{aligned}$$

PUT-CALL PARITY

The standard relationship between call and put prices is

$$C_0 + PV(K) = P_0 + S_0.$$

Including the present value of dividends D :

$$C_0 + R_f^{-\tau}X + D = P_0 + S_0.$$

FORWARD AND FUTURES PRICING

The fair forward price accounting for dividends or income is

$$F_{0\tau} = (S_0 - D)R_f^\tau.$$

For simple compounding in exercises:

$$F_0 = [S_0 - PV(\text{Dividends})] \cdot (1 + r)^T.$$

ARBITRAGE BOUNDS AND STRIKE PRICE PROPERTIES

$$\text{Lower bound for a call: } C_0 \geq \max(S_0 - D - PV(X), 0),$$

$$\text{Strike property 1: } C(K_1) \geq C(K_2) \quad \text{if } K_1 < K_2,$$

$$\text{Strike property 2: } P(K_2) - P(K_1) \leq K_2 - K_1.$$

CONTINUOUSLY COMPOUNDED PRESENT VALUE

$$PV(K) = Ke^{-rT}.$$

9.2 RATE OF RETURN AND BREAK-EVEN POINTS

BASIC RATE OF RETURN

$$\text{Rate of Return} = \frac{\text{Final Investment Value} - \text{Initial Investment Value}}{\text{Initial Investment Value}}.$$

IMPLIED CONTINUOUS RATE OF RETURN

Given PV and future value K over T periods:

$$PV \cdot e^R = K \quad \implies \quad R = \frac{\ln(K/PV)}{T}.$$

BREAK-EVEN POINTS FOR OPTIONS

Put: $S_T = \text{Strike Price} - \text{Initial Net Proceeds}$,

Call: $S_T = \text{Strike Price} + \text{Initial Net Proceeds}$.

PRESENT VALUE (SIMPLE COMPOUNDING)

For non-indexed assets or discrete compounding:

$$PV = \frac{\text{Future Value}}{(1 + r)^T}.$$

TOTAL PROFIT/LOSS OF A POSITION

Profit = Net Proceeds from Writing Options – Net Cash Outflows at Expiration.